

Research Project Report: Heart Failure Prediction AI System

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1. Introduction

This project focuses on designing and implementing a machine learning system capable of automatically classifying heart dataset records into two distinct categories: healthy and heart disease. The core methodology relies on leveraging clinical data and applying various machine learning algorithms to evaluate heart disease risk in real time.

The primary objective of this research is to compare multiple predictive models to identify the most effective architecture for data classification tasks. Furthermore, this study demonstrates how machine learning can be practically integrated into real-world applications, such as smart medical systems and advanced healthcare management.

2. Dataset Description

The dataset utilized in this empirical study was sourced from Kaggle: Heart Failure Prediction Dataset. It consists of a comprehensive collection of patient data records divided into two structural classes:

- Healthy: Representing healthy patients.
- Heart Disease: Representing patients with heart disease risks.

3. Data Preprocessing

- Label Encoding: All categorical text columns were converted to numerical values to minimize computational complexity and allow the models to focus strictly on essential structural features.
- Data Alignment: The dimensions of the features were structured into a fixed and uniform format.
- Numerical Array Conversion: The processed records were transformed into multi-dimensional numerical arrays compatible with model training pipelines.
- Data Splitting: The final dataset was split into an 80% training set and a 20% testing set to rigorously evaluate model performance on unseen data.

4. Models Implemented

- Logistic Regression: Implemented as a baseline linear model for binary classification.
- Random Forest Classifier: An ensemble learning methodology based on decision trees, utilized to improve predictive accuracy and mitigate overfitting.
- K-Nearest Neighbors (KNN): A distance-based instance classifier that categorizes sample records according to their closest training examples.
- Neural Network: A deep learning model built using the Keras framework, engineered to extract and model highly complex, non-linear patterns within the data.

5. Model Evaluation Metrics

1. Accuracy: Quantifies the overall proportion of correct predictions made by the model.
2. Precision: Measures the exactness of the model by showing how many predicted positive cases were actually correct.
3. Recall: Measures the sensitivity of the model by capturing how many actual positive cases were correctly identified.
4. F1-Score: Computes the harmonic mean of precision and recall, offering a balanced performance evaluation metric.
5. Confusion Matrix: Utilized as a visual tool to map correct and incorrect predictions across both classes.

6. What I Learned

- **Model Complexity vs. Performance:** More complex models are not always better. While the Deep Learning Neural Network demonstrated stable learning, the classic Logistic Regression and Random Forest Classifiers achieved the best overall performance and highest accuracy for this specific task.
- **The Impact of Preprocessing:** Proper data cleaning and preparation before training made the most significant difference. Converting text columns into numerical values and organizing them into a uniform structure drastically reduced computational overhead while preserving crucial features necessary for accurate classification.

7. Conclusion

- Among all tested frameworks, the traditional classifiers achieved superior overall performance and yielded the highest classification accuracy.
- The Neural Network displayed a highly stable learning trajectory and solid generalization capabilities during both the training and validation phases.
- Overall, every implemented model demonstrated robust capability on the dataset, underscoring the profound effectiveness of machine learning methodologies in handling modern classification tasks.